Option C – The Physics of Sports

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
9	(a)		Substitution into principle of moments i.e. $F\cos 8 \times 0.07 = (12.9 \times 0.19) + (74 \times 0.39)$ (1) F = 451 [N] (1) If $F\sin 7 \times 0.07$ **or** $F \times 0.07$ used correctly – award 1 mark	1	1		2	1	
	(b)		Recall $Ft = mv - mu$ or equivalent (1) Initial momentum = 907 N s and impulse = +/- 968 N s (1) Final velocity = - 0.52 [m s^{-1}] (need sign) (1) **Alternative:** Recall $F = ma$ (1) Acceleration = 47 m s^{-2} and use of $v = u + at$ (1) Final velocity = - 0.52 [m s^{-1}] (need sign) (1)	1	1 1		3	2	
	(c)	(i)	Recall drag force $F = \frac{1}{2}\rho v^2 A C_D$ (1) Density and drag coefficient are constant and increasing velocity increases the magnitude of F (1) accept density can be variable with altitude [Effective / cross-sectional / surface] area of A is less than B so drag [force] is reduced (1)	1	1 1		3		
		(ii)	Rotational kinetic energy $= \frac{1}{2}I\omega^2$ (1) ω is the number of revolutions $\times$ 2π (or ω is proportional to revs per second) so will not double (1)			2	2		

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		(iii)	Using equation for coefficient of restitution $e = \sqrt{\frac{h}{H}}$ (1) Coefficient of restitution = 0.39 (1) The same value for both surfaces (1) If no square root taken award a max of 2 marks			3	3	2	
		(iv)	Recall $T = I\alpha$; (1) Torque = 6.2 [N m] (1) **Alternative:** Recall: torque = $\frac{\text{change in angular momentum}}{\text{time}}$ (1) Torque = 6.2 [N m] (1)	1	1		2		
		(v)	Either of the vertical components calculated correctly 12sin32° = 6.4 [m s^{-1}] or 12cos32° = 10.2 [m s^{-1}] (1) Substitute values into $v^2 = u^2 + 2ax$ **or** $x = ut + ½ at^2$ (1) Time to reach height = 0.65 [s] **or** height $x = \frac{6.4^2}{2 \text{ x } 9.81}$ **or** $t = \frac{6.2}{10\cos 32}$ [= 0.61 s] (1) [from level of the thrower] Maximum height reached by the ball = 2.1 [m] [from level of the thrower] **or** height reached = 2.05 [m] (1) Overall, height = $\frac{2.1}{2.05}$ + 2.0 [= 4.1 m] so jumper will catch the ball (1)	1 1	1 1 1		5	5	
			Question 9 total	**6**	**9**	**5**	**20**	**10**	**0**